Journal of Dermatological Science 72 (2013) 116–122

Contents lists available at SciVerse ScienceDirect

Journal of Dermatological Science

journal homepage: www.jdsjournal.com

Regulation of lipid production by acetylcholine signalling in human sebaceous glands

Zheng Jun Li [a,1], Seung Bae Park [a,1], Kyung Cheol Sohn [a], Young Lee [a], Young Joon Seo [a], Chang Deok Kim [a], Youn Sung Kim [b], Jeung Hoon Lee [a], Myung Im [a,*]

[a] Department of Dermatology, College of Medicine, Chungnam National University, Daejeon, Republic of Korea
[b] The Kimbell Catholic Clinic, Daejeon, Republic of Korea

ARTICLE INFO

Article history:
Received 19 March 2013
Received in revised form 4 June 2013
Accepted 16 June 2013

Keywords:
Acetylcholine
Botulinum toxin
Lipid
Sebaceous gland
Sebocyte

ABSTRACT

Background: The extraneuronal cholinergic system has been implicated in numerous functions in the skin, such as terminal differentiation, barrier formation, sweat secretion and the microcirculation. However, the evidence for cholinergic signalling in sebaceous glands is lacking, and its role needs to be clarified.

Objective: We investigated the role of acetylcholine signalling in sebaceous glands using human sebocytes and a clinical study using botulinum toxin.

Methods: Immunohistochemistry and immunocytofluorescence were performed to evaluate cholinergic receptor levels in sebaceous glands. Lipid levels were assessed by Oil Red O staining and signalling pathways by Western blotting. To evaluate the clinical relevance, we also assessed the effect of botulinum toxin on sebum production in healthy volunteers.

Results: We demonstrated that human skin sebaceous glands *in vivo* and sebocytes *in vitro* express nicotinic acetylcholine receptor α7 (nAchRα7), and that acetylcholine increased lipid synthesis in a dose-dependent manner. When sebocytes were incubated with α-bungarotoxin, a competitive nAchR antagonist, acetylcholine failed to up-regulate lipid synthesis. Twenty healthy volunteers were enrolled in a double-blind, placebo-controlled, split-face study. A marked decrease in sebum production on the botulinum-treated side was found in volunteers with oily skin.

Conclusion: These results provide evidence that acetylcholine signalling plays a significant role in human sebaceous gland biology and identify acetylcholine signalling as a promising target in the clinical management of disorders in which sebum production is increased, such as acne vulgaris.

© 2013 Japanese Society for Investigative Dermatology. Published by Elsevier Ireland Ltd. All rights reserved.

1. Introduction

Human skin is an extension of the neuronal network that includes both adrenergic and cholinergic signal transduction pathways [1,2]. The cholinergic system has been demonstrated to influence a plethora of cutaneous cell functions, such as keratinocyte proliferation, terminal differentiation and barrier formation, and sweat secretion, as well as the microcirculation and angiogenesis [3,4].

Acetylcholine (Ach) is the cholinergic neurotransmitter and acts *via* to two principal receptor types, the nicotonic (nAchR) and the muscarinic acetylcholine receptors (mAchR), that show considerable heterogeneity in different tissues [1]. A major nAchR subunit, nAchRα7, is expressed by non-neuronal cells throughout the body, suggesting that they contribute to diverse pathophysiological processes. nAchRα7 exerts essential roles in the cholinergic anti-inflammatory pathway [5], in angiogenesis [6] and in collagen expression [7]. Additionally, there is ample evidence that nAChRα7 plays an important role in keratinocyte adhesion, migration, differentiation and apoptosis [1,4,8].

Interestingly, the sebaceous gland is not innervated by the parasympathetic nervous system, but was recently proved to prominently express distinct nAchR subunits, including nAchRα7, and releases Ach by autocrine mechanisms [9,10]. The regulated expression of AchR in sebaceous glands advocates a role for Ach in sebum production and as a promoter of sebocyte differentiation, thus offering an explanation of skin diseases associated with altered sebum production after chronic nicotine exposure. However, no available data show that Ach signalling directly influences lipid production. The mechanism of action of Ach in human sebaceous glands and its potential efficacy also need to be clarified.

* Corresponding author at: Department of Dermatology, Chungnam National University Hospital, 282 Munhwa-ro, Jung-Gu, Daejeon 301-721, Republic of Korea. Tel.: +82 42 280 7702; fax: +82 42 280 7932.
E-mail address: im1177@hanmail.net (M. Im).
[1] The first two authors contributed equally to this work.

0923-1811/$36.00 © 2013 Japanese Society for Investigative Dermatology. Published by Elsevier Ireland Ltd. All rights reserved.
http://dx.doi.org/10.1016/j.jdermsci.2013.06.009

In this study, we investigated the effect of Ach on sebum production in human sebaceous glands *in vivo* and *in vitro*. Our findings provide compelling evidence that Ach signalling potently stimulates sebum production. It is thus tempting to propose that agents that block cholinergic signalling, such as botulinum toxin, could be interesting candidates for the treatment of skin disorders with increased sebum production, such as acne vulgaris.

2. Materials and methods

2.1. Specimens

Specimens for sebocyte culture and immunohistochemistry were obtained surgically from the faces of patients undergoing plastic surgery. All human skin samples were obtained with the written informed consent of the donors and in accordance with the ethical committee approval process of the Institutional Review Board of Chungnam National University Hospital.

2.2. Human sebocyte culture

Primary cultures of sebocytes were established according to a method described previously [11]. Briefly, sebaceous glands were isolated from separated epidermis under a microscope, and transferred to a tissue culture dish. The cells were maintained in DMEM/Ham's F12 medium (Invitrogen, Carlsbad, CA, USA) supplemented with 10% foetal bovine serum (Gibco BRL, Rockville, MD, USA) and 5 ng/ml recombinant human epidermal growth factor (Invitrogen) at 37 °C in a humidified atmosphere containing 5% CO_2. After the cells became sub-confluent, they were harvested with 0.05% Trypsin-EDTA (Gibco BRL) and sub-cultured. Cells obtained after the second passage were used in this study.

2.3. Immunohistochemistry

Tissue samples were fixed with 10% formaldehyde, embedded in paraffin, and sectioned to a thickness of 4 μm. The sections were deparaffinised in xylene and then rehydrated through an alcohol series. Antigens were retrieved by pressure-cooking the sections in citrate buffer (10 mM citric acid, pH 6.0) for 4 min. Endogenous peroxidase activity was blocked by incubating the sections in 2% hydrogen peroxide for 30 min at room temperature. The primary antibody was anti-nAChRα7 (Abcam, Cambridge, UK), diluted 1:200. The sections were then incubated with anti-rabbit secondary antibody at room temperature for 30 min. The sections were finally incubated with diaminobenzidine tetrachloride solution at room temperature for 5 min, and counterstained with Mayer's haematoxylin.

2.4. Immunocytofluorescence

Primary sebocytes were grown on coverslips, fixed by incubation with 4% paraformaldehyde for 20 min, and permeabilised by incubation with 0.1% Triton X-100 in PBS for 10 min at room temperature. Cells were then incubated overnight at 4 °C with primary antibody (anti-nAChRα7) (Abcam) and then for 1 h at room temperature with FITC-conjugated secondary antibody, and were finally visualised under a fluorescence microscope (Olympus Corporation, Tokyo, Japan).

2.5. Western blotting

Cells were lysed in Proprep solution (Intron, Daejeon, Korea). Samples were resolved in SDS-polyacrylamide gels, transferred onto nitrocellulose membranes, and incubated with appropriate antibodies. Blots were then incubated with peroxidase-conjugated secondary antibodies and visualised by enhanced chemiluminescence (Intron). Primary antibodies against the following proteins were used: nAChRα7 (Abcam), extracellular signal-regulated kinase (ERK), phospho-ERK, actin (Santa Cruz Biotechnology, Santa Cruz, CA, USA) and peroxisome proliferator-activated receptor-γ (PPAR-γ) (Cell Signalling Technology, Danvers, MA, USA).

2.6. RT-PCR

To evaluate gene expression, total RNA was extracted using an easy-BLUE RNA extraction kit (Intron). Total RNA (2 μg) was reverse-transcribed using M-MLV reverse transcriptase (ELPIS Biotech, Daejeon, Korea). Aliquots of an RT mixture were subjected to PCR using specific primer sets. The following primers were used: nAChRα7, forward 5′-GCACCTCATGCATGGTACAC-3′ and reverse 5′-GGACACAGCCTCCACAAAGT-3′; nAChRα10, forward 5′-CTCTCAAGCTGTTCCGTGACC-3′ and reverse 5′-AAGGCTGCTACATC-CACGC-3′; mAChRm2, forward 5′-CCAAGTCTGGTACAAGGA AG-3′ and reverse 5′-TCCACCGTGGTATTAGTTG-3′; mAChRm4, forward 5′-CGGTGCTGTACATCCACATCT-3′ and reverse 5′-CTCTGTGGTGGA-CAGCTCTGTG-3′; PPAR-γ, forward 5′-TCTCTCCGTAATGGAAGACC-3′ and reverse 5′-GCATTATGAGACATCCCCAC-3′; and cyclophilin (control), forward 5′-CTCCTTTGAGCTGTTTGCAG-3′ and reverse 5′-CACCACATGCTTGCCATCCA-3′.

2.7. Oil Red O staining

Primary sebocytes were seeded on 12-well culture plates, incubated overnight, and then treated with various doses of Ach for 24 h. At the end of the treatment period, cells were washed with PBS and fixed by incubation in 10% formaldehyde for 5 min. Fixed cells were stained for 10 min with filtered Oil Red O working solution, prepared immediately before use by making a 6:4 mixture of stock (0.5% Oil Red O (Sigma, St. Louis, MO, USA) in 99% isopropanol) and distilled H_2O. Stained cells were washed in distilled H_2O, counterstained with haematoxylin, and visualised by microscopy.

2.8. Lipid detection

For quantitative detection of intracellular lipids, Oil Red O was removed by incubating the cells with 100% isopropanol for 10 min. After pipetting several times to ensure that all Oil Red O had been eluted, supernatant Oil Red O levels were determined by measuring the optical density (OD) at 500 nm. To calculate the amount of lipid per cell, the OD value was normalised to the cell count, determined using a FACSCanto™ II flow cytometer and FACSDiva software (BD Biosciences, San Jose, CA, USA).

2.9. Construction of a recombinant adenovirus expressing microRNA specific for ERK1/2

Three pre-microRNA (miR) sequences that targeted the 3′-untranslated region of the human ERK1/2 mRNA were designed using an online tool (RNAi Designer, Invitrogen). Double-stranded DNA oligonucleotides corresponding to ERK1/2-specific pre-microRNAs and a control sequence were individually cloned into the parental vector pcDNA6.2-GW/miR (Invitrogen). Recombinant adenovirus was produced using the ViraPower Adenoviral Expression System (Invitrogen), according to the manufacturer's instructions.

2.10. Clinical study of botulinum toxin

Twenty healthy volunteers (6 men, 14 women) aged 21–37 years were enrolled in this double-blind, placebo-controlled, split-face

study after signing an informed consent form for this off-label use of botulinum toxin, in accordance with the ethical committee approval process of Chungnam National University Hospital. All volunteers had no accompanying diseases and volunteers who had a history of cosmetic procedures or retinoid treatment within the past 6 months were excluded from this study. Facial sebum secretion was measured using a Sebumeter (SM 815®, CL-Electronics, Cologne, Germany) at four different sites on each cheek. Skin types were classified based on the sebum measurements according to the manufacturer's guidelines, as follows: $>180\ \mu g/cm^2$, oily type; $\leq 180\ \mu g/cm^2$, normal to dry type. All procedures were performed by the same investigator. One side of face was injected intradermally with botulinum toxin A (Meditoxin®, Medytox, Seoul, Korea) injections and the other side with normal saline (as a control). The treatment areas were staggered each 4 points at 1 cm intervals with an injection volume of 0.1 ml (2 U/0.1 ml) per treatment site. Patients were instructed to use non-comodogenic makeup and skin products after therapy. The participants were followed up 1 and 4 weeks after the injections. To avoid diurnal variation, sebum levels were measured at 10 am. Participants were asked not to put on any cosmetics for 2 h before the measurements. A constant temperature ($20 \pm 1\ ^\circ C$) and humidity ($40 \pm 2\%$) were maintained during sebum measurement. Additionally, pore sizes were measured at follow-up. The overall area covered by follicular orifices was calculated by digital image analysis, performed using a phototrichogram system (Folliscope, Lead-M Co., Seoul, Korea).

2.11. Statistical analysis

All experiments were repeated at least three times with different batches of cells. Data were evaluated statistically using Student's *t*-test. Statistical significance was set at $P < 0.05$.

3. Results

3.1. nAChRα7 is expressed in human sebaceous glands and primary sebocytes

As the preliminary study, we checked 4 subtypes of AchR, such as α7, α10, β2 and β4 that were known to be up-regulated AchR in mature sebocytes as compared with undifferentiated sebocytes in the previous report [9]. As a result of RT-PCR in human sebocytes, the mRNA expression of nAchRα7 was markedly higher than other 3 subtypes of AchR (Fig. 1A). Therefore, we decided to investigate

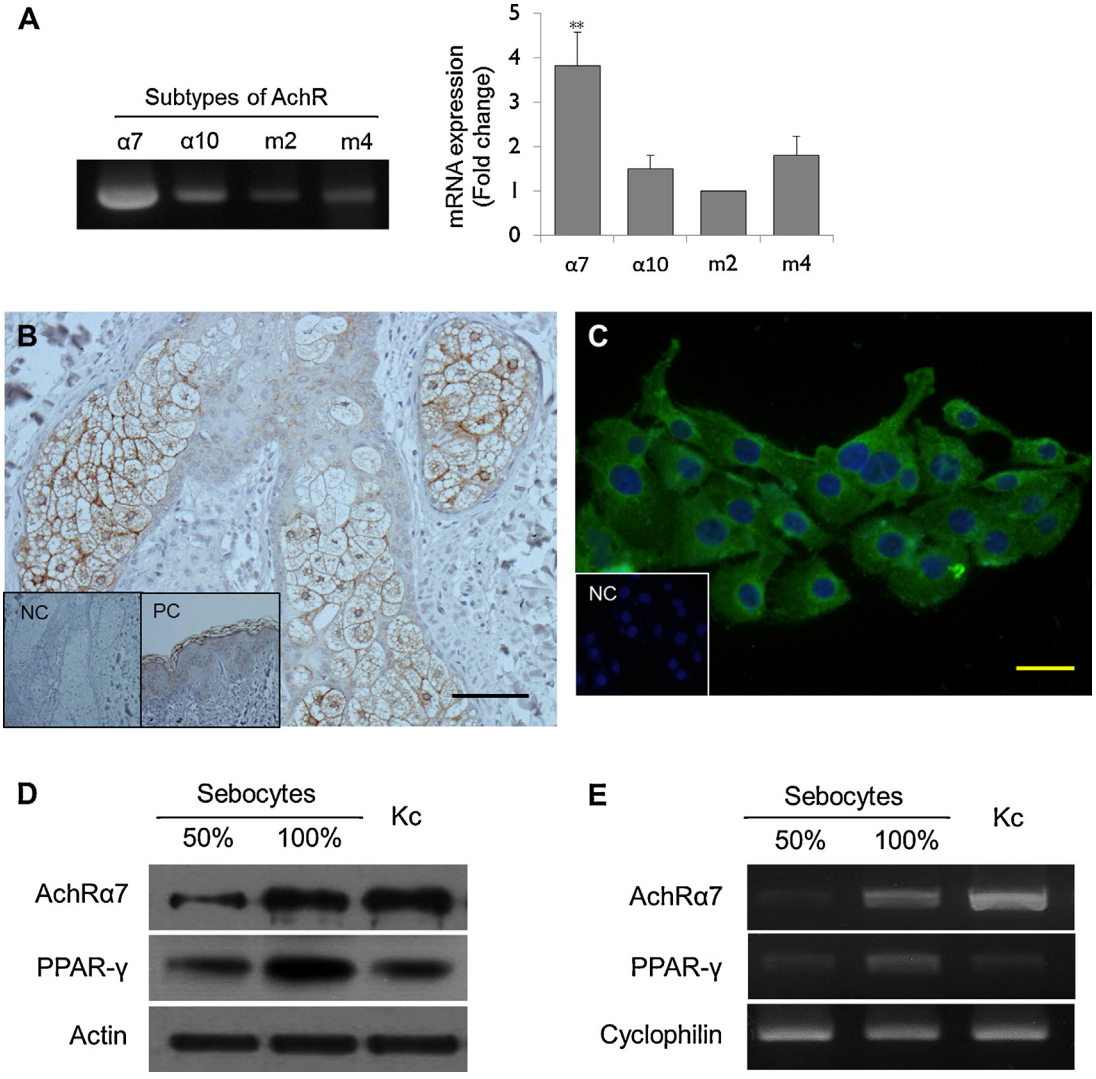


Fig. 1. Expression of nAChRα7 in human sebaceous gland and sebocytes. (A) Amplification of 4 types of AChR mRNA from human primary sebocytes. (B) Human sebaceous glands were assayed for nAChRα7 by immunohistochemistry. Scale bars = 100 μm. (C) Immunofluorescence labelling of nAChRα7 (green) in human sebocytes. Nuclei were counterstained with DAPI (blue). Scale bars = 20 μm. (D, E) Western blot and RT-PCR of lysates of sebocytes harvested at various confluence levels. Human differentiated keratinocytes overexpressing nAChRα7 were used as positive controls. $^{**}P < 0.01$ *vs.* nAChRα10, mAChRm2 and m4. NC, preabsorption negative control. PC, positive controlled epidermis. Kc, keratinocytes.

the role of nAChRα7 in human sebaceous glands. Immunohistochemical analysis also revealed the presence of nAChRα7 in normal human sebaceous glands. Moreover, we found that nAChRα7-specific immunosignals were more prominent in more differentiated sebaceous gland cells, similar to earlier reports (Fig. 1B) [9]. We then assessed the presence of nAChRα7 in human sebocytes. Using various immunocytochemical methods, Western blotting and RT-PCR, we obtained evidence that human sebocytes also express nAChRα7 at the protein and mRNA levels (Fig. 1C and E). As a positive control for these experiments, the nAChRα7 receptor was successfully probed in normal human epidermal keratinocytes, which have been reported to express this receptor [4,8,9]. Interestingly, similar to immunohistochemical analysis of tissues, levels of nAChRα7 protein and mRNA were markedly increased in more confluent cultures, which were possibly more differentiated (Fig. 1D and E). These results supported our hypothesis that the differentiation of sebocytes is affected by the nAChRα7 expression level.

3.2. Activation of nAChRα7 by Ach induces lipid synthesis in human sebocytes

To investigate the role of cholinergic signalling in the human sebaceous gland, primary sebocytes were treated with Ach. When sebocytes were treated with 10 nM Ach, an increase in lipid accumulation in the cytoplasm was detected by microscopy after Oil Red O staining (Fig. 2A). When sebocytes were treated with various doses of Ach, lipid synthesis in sebocytes increased in a dose-dependent manner (Fig. 2B). We then determined whether this increase in lipid synthesis in sebocytes could be inhibited using nAChRα7 antagonists. α-Bungarotoxin, a pharmacologic antagonist of nAChRα7, suppressed Ach-induced lipid synthesis in sebocytes (Fig. 2C).

3.3. Ach enhances lipid synthesis via ERK signalling

These findings prompted us to investigate the molecular mechanism by which cholinergic signalling enhances lipid synthesis. Previous studies showed that nAChRα7 stimulation indirectly triggers the activation of an intracellular MAPK/ERK pathway in various cell types [12,13]. We therefore examined the effects of Ach on the phosphorylation of ERK in sebocytes by Western blotting. Levels of phosphorylated ERK were increased by Ach, a response that was blocked by α-bungarotoxin. Furthermore, levels of PPAR-γ, a potential modulator of lipid production in human sebocytes [14], showed results similar to those for phosphorylated ERK (Fig 3A). To determine the mechanism more precisely, we generated a recombinant adenovirus expressing microRNA specific for ERK2 (miERK) to knock-down expression of this gene. As shown in Fig. 3B, miERK significantly reduced the protein level, whereas a scrambled microRNA did not. Regarding the relationship between ERK1/2 and PPAR-γ, we found that downregulation of ERK1/2 resulted in a decrease in PPAR-γ signalling. And these findings were confirmed by Oil Red O staining that showing the reduction of lipid synthesis by miERK (Fig. 3C and D). These results suggest that cholinergic signalling increases lipid synthesis in human sebocytes *via* activation of ERK signalling and that this

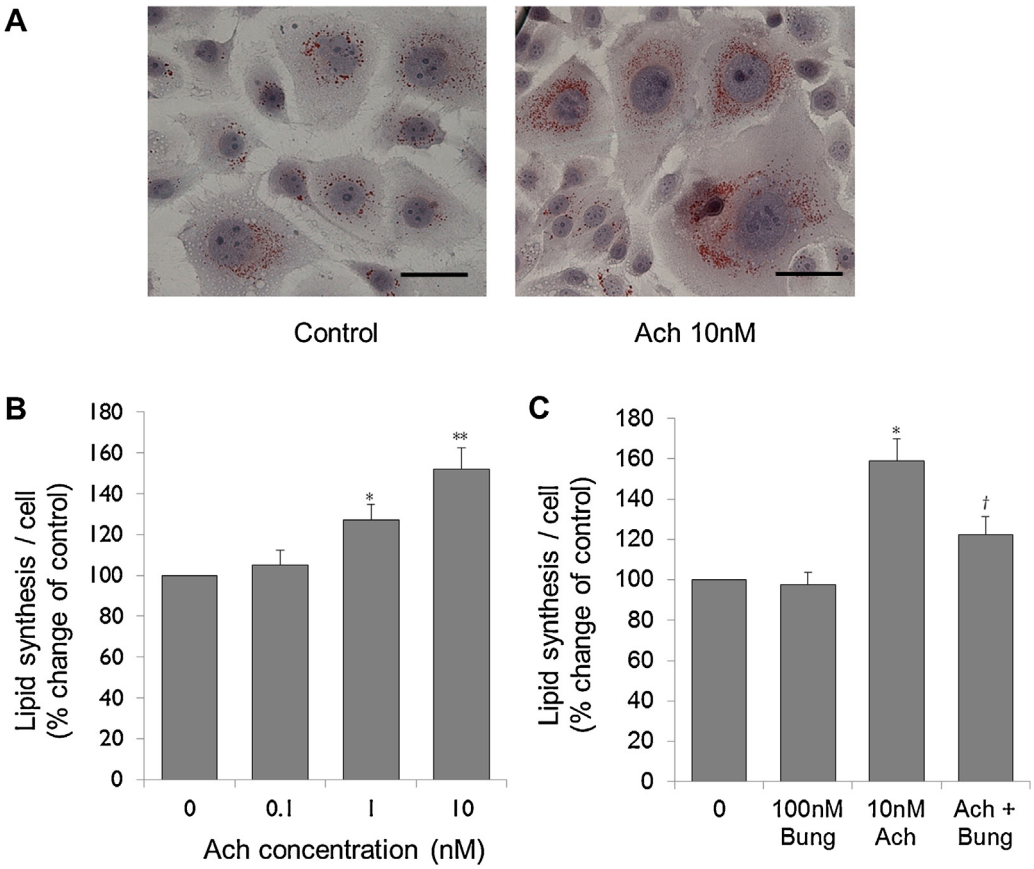


Fig. 2. Increased lipid synthesis in sebocytes in response to Ach. (A) Detection of intracellular lipids in sebocytes treated with 10 nM Ach by microscopy after Oil Red O staining. Scale bars = 20 μm. (B) Lipid levels in sebocytes treated with various doses of Ach, calculated as percentages of the value of untreated cells. (C) Sebocytes were stimulated for 24 h with 10 nM Ach in the presence or absence of 100 nM α-bungarotoxin. Intracellular lipid levels were then measured. Data are means ± SE ($n = 5$). $^{*}P < 0.05$, $^{**}P < 0.01$ *vs.* untreated cells; $^{\dagger}P < 0.05$ *vs.* Ach-treated cells. Bung, α-bungarotoxin.

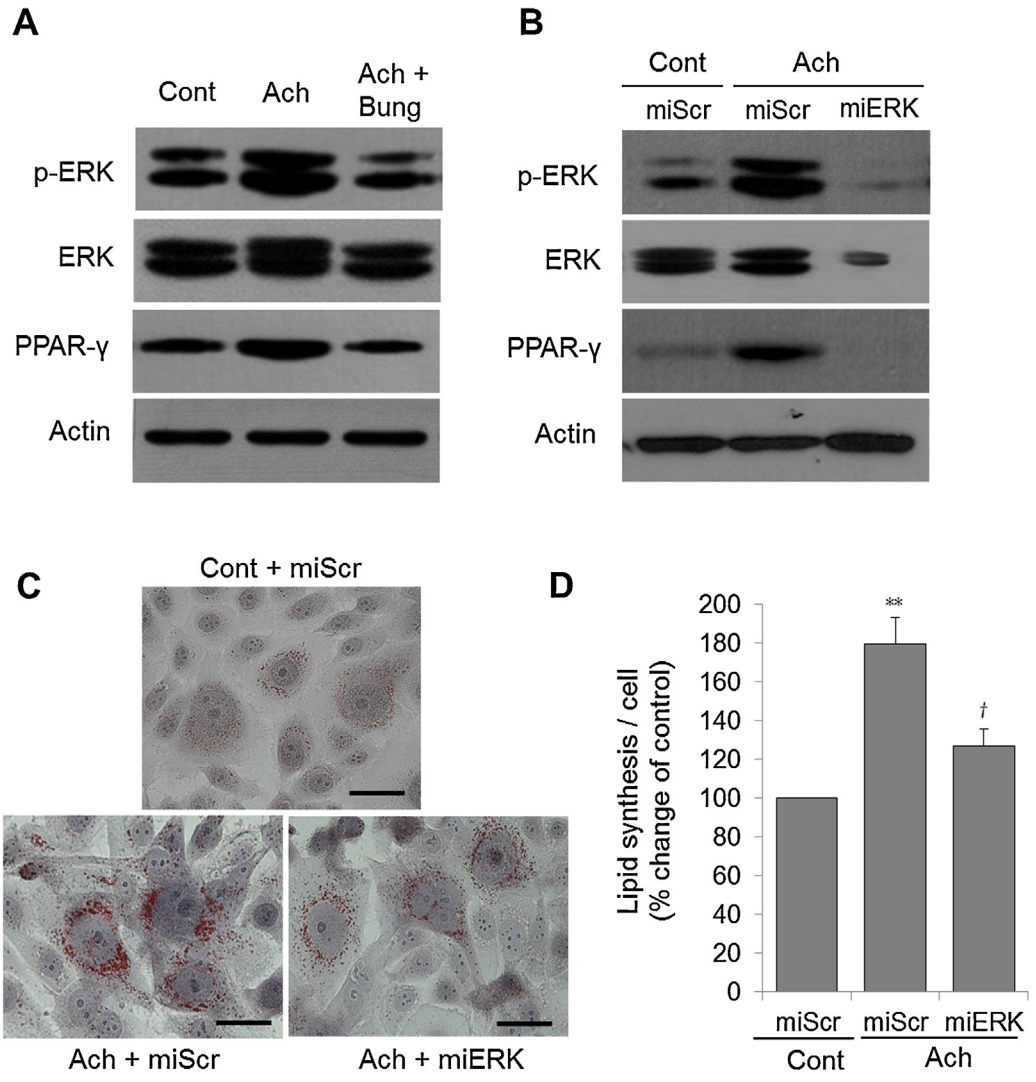


Fig. 3. Regulation of lipid synthesis in Ach treated-sebocytes *via* ERK signalling. (A) Sebocytes were treated with 10 nM Ach for 24 h in the absence or presence of 100 nM α-bungarotoxin. Whole-cell lysates were prepared and subjected to Western blotting. Blots were incubated with antibodies specific for total and phosphorylated ERK and PPAR-γ. (B) Sebocytes were transduced with an adenovirus expressing microRNA specific for ERK 1/2 (miERK) at a multiplicity of infection of 10 for 6 h. An adenovirus expressing scrambled microRNA (miScr) was used as the negative control. (C) Sebocytes transducing with miScr or miERK were treated for 24 h with 10 nM Ach. Detection of intracellular lipids in sebocytes by microscopy after Oil Red O staining. Scale bars = 20 μm. (D) Lipid levels in sebocytes calculated as percentages of the value of control cells. Data are means ± SE ($n = 5$). $^{**}P < 0.01$ *vs.* Cont + miScr; $^{\dagger}P < 0.05$ *vs.* Ach + miScr. Cont, vehicle-treated control.

suppression may correlate with inhibition of PPAR-γ-induced lipogenesis in sebocytes.

3.4. Botulinum toxin decreases sebum production in oily skin

To investigate the role of cholinergic signalling in the human sebaceous gland *in vivo*, we assessed the effect of botulinum toxin on sebum production. Ten of the 20 participants were classified as having oily skin, and the others dry to normal skin. In the group with dry-to-normal skin, there was no significant difference between the skin on both sides of the face 1 and 4 weeks after injection, although a global improvement pattern was evident. In contrast, in the group with oily skin, a marked decrease in sebum production on the botulinum-treated side compared to the saline-injected side was identified, particularly at 4 weeks after injection (Fig. 4A). Next, we evaluated pore size changes induced by botulinum toxin, because a previous study demonstrated that sebum output contributes to the sizes of facial pores [15]. A statistically significant decrease in pore size 4 weeks after injection in the oily group was found; there was no difference in the dry to normal group (Fig. 4B). None of the enrolled subjects experienced significant adverse effects, such as allergic reaction, facial palsy or severe paralysis of muscles adjacent to the point of injection, either during or after the study. A mild-to-moderate stinging sensation was noted.

4. Discussion

A complex network of cellular and biochemical mechanisms facilitates the pathogenesis of skin disease upon activation of the stress response through cholinergic signalling [4,10]. Patients with Atopic dermatitis (AD) have a significant increase in Ach expression in the skin compared to normal individuals [16]. Moreover, Ach contributes to increased pruritus of AD due to its ability to cause vasodilatation and the wheal and flare reaction [16,17]. The clinical onset and exacerbation of psoriasis are associated with increased corticotrophin-releasing hormone level that is dependent on cholinergic activation [18,19]. In the pathogenesis of hidradenitis suppurativa (HS), smoking seems to be the most important factor [20,21]. There are also data

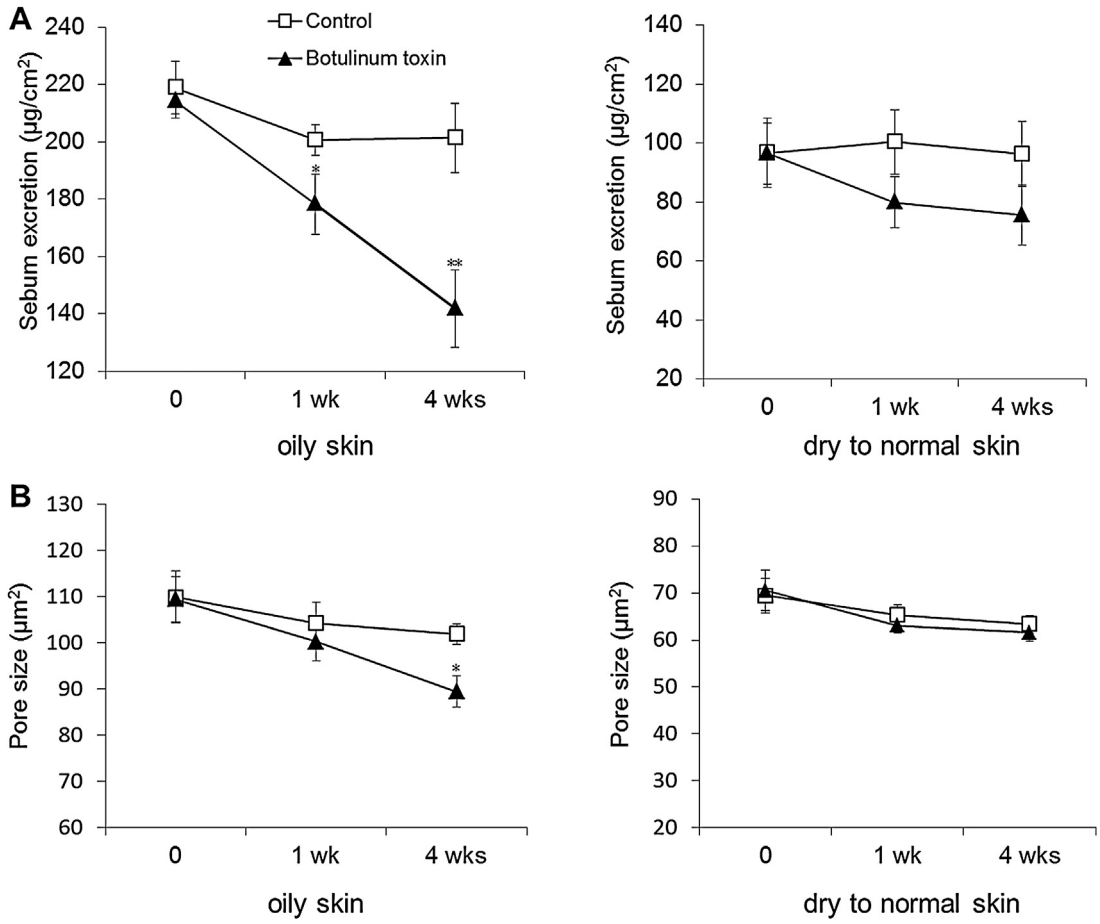


Fig. 4. Effect of botulinum toxin on sebum production. Changes in mean sebum output level ($\mu g/cm^2$) (A) and pore size (μm^2) (B) after botulinum toxin injection in the oily and dry-to-normal groups. Data are means ± SE ($n = 10$). $^*P < 0.05$, $^{**}P < 0.01$ *vs.* before treatment. The English in this document has been checked by at least two professional editors, both native speakers of English. For a certificate, please see: http://www.textcheck.com/certificate/F7irKI.

showing that nicotine and the non-neuronal cholinergic system play a causative role in the pathogenesis of HS by promoting infundibular epithelial hyperplasia [22].

Exacerbation of acne is known to result from emotional stress, and there are several lines of evidence for the participation of cholinergic signalling in the pathogenesis of acne. Although the correlation between acne and smoking remains controversial, several clinical studies showed a positive correlation between smoking and acne [23–26]. Nicotine, a major component of tobacco, may exert its disease-provoking effects by promoting infundibular epithelial hyperplasia and thus follicular plugging through direct action on nAchR present in acne lesions. However, other aspects of cholinergic signalling affect sebum production in the pathogenesis of acne. One is the differentiation-specific expression of different nAchR and mAchR receptors in sebaceous glands [9], and the other is that smokers affected by acne show a significant increase in sebum secretion in comparison with non-smokers [27]. Based on these observations, we hypothesised that cholinergic signalling is directly correlated with sebum production and investigated the central role of cholinergic signalling in lipid production in sebaceous glands using *in vivo* and *in vitro* methods.

The functional data reported here identify Ach as a significant new player in human sebocyte biology, specifically sebum production. This conclusion is supported by nAchRα7 expression in human sebocytes and the inhibitory effect of α-bungarotoxin on Ach-induced lipid synthesis. We next investigated the mechanism by which Ach signalling increases sebum production. Ach has been shown to modulate neuronal and extra-neuronal differentiation during early development. However, the intracellular mechanisms underlying these effects of AchR signalling are not fully understood. Intriguingly, activation of diverse signalling molecules, such as Ras-mitogen-activated protein kinase, phosphatidylinositol 3-kinase-Akt, protein kinase C and c-Src is modulated by AChRs [28,13]. We showed that Ach increased lipid synthesis *via* an MAPK/ERK pathway related to up-regulation of PPAR-r. There is some controversy as to whether ERK signalling regulates PPAR-r positively or negatively. In previous studies, activation of ERK was associated with a reduction in PPAR-r transcriptional activity [29,30]. However, our data suggest that activation of ERK signalling enhances PPAR-r activity [31,32]. Therefore, we sought to determine the nature of the relationship between ERK and PPAR-r in human sebocytes by knockdown of ERK using microRNA. Finally, we showed that the suppression of ERK signalling was associated with a decrease in PPAR-γ and proposed a possible mechanism by which Ach increases lipid synthesis by enhancing ERK and PPAR-r levels in sebocytes.

We next carried out an *in vivo* clinical study using botulinum toxin to confirm this *in vitro* experimental evidence. Botulinum toxin A is a powerful neurotoxin that acts by blocking the release of acetylcholine, and has been used in the treatment of muscle spasm disorders, pain and hyperhidrosis [33]. Currently, many off-label cosmetic applications of botulinum toxin are under evaluation, therefore, we carried out a double-blind, placebo-controlled, split-face study with 20 volunteers. Botulinum toxin effectively reduced sebum production and pore size in the oily skin group, but had no effect in the dry-to-normal skin group. This finding supports the notion that sebum production is influenced by cholinergic

signalling and that botulinum toxin decreases sebum production by interfering with cholinergic transmission between autonomic nerve terminals and sebaceous glands. Additionally, we suggest that oily skin may be more susceptible to cholinergic regulation than normal skin due to the greater abundance of mature sebocytes with more cholinergic receptors, as demonstrated by our *in vitro* findings. These findings also could be affected by other factors including sweat secretion, barrier formation, collagen synthesis and so on, which may also be affected by botulinum toxin injection. Therefore, it needs to further clinical investigation to exclude the other factors affecting the sebum secretion.

In conclusion, we have demonstrated that cholinergic receptors are expressed in human sebocytes and that acetylcholine induced lipid synthesis in human sebocytes *via* the ERK pathway. Moreover, blockade of cholinergic signalling suppressed acetylcholine-induced sebum production. Collectively, these findings support the concept that acetylcholine is a significant player in human sebocyte biology and identify cholinergic signalling as a promising target in the clinical management of hyperseborrhoeic states, such as acne vulgaris.

Funding

There was no funding source.

References

[1] Grando SA, Zelickson DB, Kist DA, Weinshenker D, Bigliardi PL, Wendelschafer-Crabb G, et al. Keratinocyte muscarinic acetylcholine receptors. Immunolocalization and partial characterization. J Invest Dermatol 1995;104:95–100.
[2] Schallreuter KU, Lemke KR, Pittelkow MR, Wood JM, Körner C, Malik R. Catecholamines in human keratinocyte differentiation. J Invest Dermatol 1995;104:953–7.
[3] Kurzen H, Schallreuter KU. Novel aspects in cutaneous biology of acetylcholine synthesis and acetylcholine receptors. Exp Dermatol 2004;13:27–30.
[4] Curtis BJ, Radek KA. Cholinergic regulation of keratinocyte innate immunity and permeability barrier integrity: new perspectives in epidermal immunity and disease. J Invest Dermatol 2012;132:28–42.
[5] Metz CN, Tracey KJ. It takes nerve to dampen inflammation. Nat Immunol 2005;6:756–7.
[6] Cooke JP. Angiogenesis and the role of the endothelial nicotinic acetylcholine receptor. Life Sci 2007;80:2347–51.
[7] Sekhon HS, Keller JA, Proskocil BJ, Martin EL, Spindel ER. Maternal nicotine exposure upregulates collagen gene expression in fetal monkey lung. Association with alpha7 nicotinic acetylcholine receptors. Am J Respir Cell Mol Biol 2002;26:31–41.
[8] Arredondo J, Nguyen VT, Chernyavsky AI, Bercovich D, Orr-Urtreger A, Kummer. et al. Central role of alpha7 nicotinic receptor in differentiation of the stratified squamous epithelium. J Cell Biol 2002;159:325–36.
[9] Kurzen H, Berger H, Jäger C, Hartschuh W, Näher H, Gratchev A, et al. Phenotypical and molecular profiling of the extraneuronal cholinergic system of the skin. J Invest Dermatol 2004;125:937–49.
[10] Kurzen H, Wessler I, Kirkpatrick CJ, et al. The non-neuronal cholinergic system of human skin. Horm Metab Res 2007;39:125–35.
[11] Xia L, Zouboulis CC, Detmar M, Kawashima K, Grando SA. Isolation of human sebaceous glands and cultivation of sebaceous gland-derived cells as an in vitro model. J Invest Dermatol 1989;93:315–21.
[12] Arias HR, Richards VE, Ng D, Ghafoori ME, Le V, Mousa SA. Role of non-neuronal nicotinic acetylcholine receptors in angiogenesis. Int J Biochem Cell Biol 2009;41:14441–51.
[13] Resende RR, Adhikari A. Cholinergic receptor pathways involved in apoptosis, cell proliferation and neuronal differentiation. Cell Commun Signal 2009;27:7–20.
[14] Downie MMT, Sanders DA, Maier LM, Stock DM, Kealey T. PPAR and FXR ligands differentially regulate sebaceous differentiation in human sebaceous gland organ cultures in vitro. Br J Dermatol 2004;151:766–75.
[15] Roh M, Han M, Kim D, Chung K. Sebum output as a factor contributing to the size of facial pores. Br J Dermatol 2006;155:890–4.
[16] Wessler I, Reinheimer T, Kilbinger H, Bittinger F, Kirkpatrick CJ, Saloga J, et al. Increased acetylcholine levels in skin biopsies of patients with atopic dermatitis. Life Sci 2003;72:2169–72.
[17] Rukwied R, Heyer G. Administration of acetylcholine and vasoactive intestinal polypeptide to atopic eczema patients. Exp Dermatol 1999;8:39–45.
[18] Kim JE, Cho DH, Kim HS, Kim HJ, Lee JY, Cho BK, et al. Expression of the corticotropinreleasing hormone-proopiomelanocortin axis in the various clinical types of psoriasis. Exp Dermatol 2007;16:104–9.
[19] Karanth S, Lyson K, McCann SM. Effects of cholinergic agonists and antagonists on interleukin-2-induced corticotropin-releasing hormone release from the mediobasal hypothalamus. Neuroimmunomodulation 1999;6:168–74.
[20] Sellheyer K, Krahl D. Hidradenitis suppurativa is acne inversa! An appeal to (finally) abandon a misnomer. Int J Dermatol 2005;44:535–40.
[21] Jansen I, Altmeyer P, Plewig G. Acne inversa (alias hidradenitis suppurativa). J Eur Acad Dermatol Venereol 2001;15:532–40.
[22] Hana A, Booken D, Henrich C, Gratchev A, Maas-Szabowski N, Goerdt S, et al. Functional significance of non-neuronal acetylcholine in skin epithelia. Life Sci 2007;80:2214–20.
[23] Goulden V, Clark SM, Cunliffe WJ. Post-adolescent acne: a review of clinical features. Br J Dermatol 1997;136:66–70.
[24] Schäfer T, Nienhaus A, Vieluf D, Berger J, Ring J. Epidemiology of acne in general population: the risk of smoking. Br J Dermatol 2001;145:100–4.
[25] Klaz I, Kochba I, Shohat T, Zarka S, Brenner S. Severe acne vulgaris and tobacco smoking in young men. J Invest Dermatol 2006;126:1749–52.
[26] Capitanio B, Sinagra JL, Ottaviani M, Bordignon V, Amantea A, Picardo M. 'Smoker's acne': a new clinical entity. Br J Dermatol 2007;157:1070–1.
[27] Capitanio B, Sinagra JL, Ottaviani M, Bordignon V, Amantea A, Picardo M. Acne and smoking. Dermatoendocrinol 2009;1:129–35.
[28] Marte BM, Downward J. PKB/Akt: connecting phosphoinositide 3-kinase to cell survival and beyond. Trends Biochem Sci 1997;22:355–8.
[29] Hu E, Kim JB, Sarraf P, Spiegelman BM. Inhibition of adipogenesis through MAP kinase-mediated phosphorylation of PPAR. Science 1996;274:2100–3.
[30] Camp HS, Tafuri SR. Regulation of peroxisome proliferator activated receptor activity by mitogen activated protein kinase. J Biol Chem 1997;272:10811–16.
[31] Prusty D, Park BH, Davis KE, Spiegelman BM. Activation of MEK/ERK signaling promotes adipogenesis by enhancing peroxisome proliferator-activated receptor gamma (PPARgamma) and C/EBPalpha gene expression during the differentiation of 3T3-L1 preadipocytes. J Biol Chem 2002;277:46226–32.
[32] Chana RS, Lewington AJ, Brunskill NJ. Differential effects of peroxisome proliferator activated receptor-gamma (PPAR gamma) ligands in proximal tubular cells: thiazolidinediones are partial PPAR gamma agonists. Kidney Int 2004;65:2081–90.
[33] Jankovic J. Botulinum toxin in clinical practice. J Neurol Neurosurg Psychiatry 2004;75:951–7.